

One person, two possible paths

A lesson on causal effects across time and people

Hector receives treatment and improves. What would have happened without it? That missing comparison is the starting point. A survival study then asks what can be learned about an *average* effect, at a specified time, for a specified population.

Learning goals

By the end, learners should be able to distinguish a before–after change from a causal contrast; name a survival estimand and its assumptions; interpret horizon estimates and a fitted trajectory; and explain why a conditional average is not an observed individual treatment effect.

A suggested class sequence

Minutes	Activity
0–10	Describe Hector's missing outcome. Ask which comparison a before–after plot actually makes.
10–25	Define the survival snapshot and separate individual, sample and population effects.
25–40	Discuss treatment assignment, censoring and overlap. Identify what adjustment must assume.
40–55	Explore the five horizons. Compare the CSF points, CAST curve and simulated answer key.
55–70	Run or inspect the small R lab. Use the frozen results if class time or computing access is limited.
70–90	Optional: critique the fit, discuss heterogeneity and write a causal roadmap for a proposed study.

Before class

Learners need basic probability and a distinction between association and causation. R is optional. The kit includes editable slides, code, a synthetic cohort and separate answer key. Its README gives run and compilation commands.

The question comes before the method

Define the population, baseline strategies, outcome and follow-up. State how the data and assumptions connect to that question. Then estimate and interpret with uncertainty. In this teaching adaptation of the causal roadmap, CSF and CAST belong at the estimation step.

Roadmap: Petersen & van der Laan (2014), [doi:10.1097/EDE.0000000000000078](https://doi.org/10.1097/EDE.0000000000000078). Implementation authority: [Igor Shuryak's demo repository](#).

1 The missing outcome and the snapshot

Potential outcomes describe what would occur under specified alternative treatment choices. Let $W_i \in \{0, 1\}$ be person i 's baseline treatment and $T_i(w)$ their potential event time under choice w . At horizon t , define

$$A_i^w(t) = \mathbf{1}\{T_i(w) > t\}, \quad \delta_i(t) = A_i^1(t) - A_i^0(t).$$

$A_i^w(t)$ is a survival indicator: 1 means alive beyond t , 0 means the event has occurred by t . The individual effect $\delta_i(t)$ is -1 , 0 or 1. We cannot observe both potential outcomes for the same person at the same horizon. Randomization does not fill in that missing value.

Under **consistency**, the event time on the assigned path is $T_i = T_i(W_i)$. An observed improvement before versus after treatment compares different times on that one path. It does not compare treatment choices at the same time.

Three averages that should not be conflated

Realized sample	$\Delta_{\text{SATE}}(t) = \frac{1}{n} \sum_{i=1}^n [A_i^1(t) - A_i^0(t)]$ The average of the realized, paired potential outcomes in this sample; some are unobserved.
Population	$\Delta(t) = \mathbb{P}\{T(1) > t\} - \mathbb{P}\{T(0) > t\}$ The survival-probability difference in a stated target population.
Conditional	$\tau(t, x) = \mathbb{E}[A^1(t) - A^0(t) \mid X = x]$ The average contrast for people with baseline profile x; it is not $\delta_i(t)$.

What the simulation's answer key means

The generator supplies conditional survival probabilities $S_w(t \mid x)$. The answer key averages their contrast over the generated baseline profiles:

$$\Delta_{\text{oracle}, X}(t) = \frac{1}{n} \sum_{i=1}^n \{S_1(t \mid X_i) - S_0(t \mid X_i)\}.$$

This is a sample-standardized *probability* contrast, also interpretable as the expected SATE given these profiles under the generating model. It is not the realized SATE and does not reveal whose outcome treatment changed. For simulations with a hidden baseline factor, the oracle also uses that factor; the fitted models do not. The supplied small cohort has no hidden confounding.

Read the units before reading the direction

A difference of 0.10 is **10 percentage points** (pp). It is neither a 10% relative increase nor a hazard ratio. Positive values favor treatment on this survival scale. Hector's opening sketch uses symptom burden, where lower is better; that vertical direction does not carry over to survival.

Potential outcomes and average effects: Hernán & Robins, [Causal Inference: What If](#), chapters 1–3.

2 What makes a comparison causal?

Confounding occurs when treatment groups differ in causes of the outcome. In this simulation, baseline health affects both treatment assignment and survival. Comparing the observed groups without adjustment can therefore mix treatment effects with pre-existing differences.

Observed survival data

Let C_i be a censoring time. The recorded data include baseline covariates X_i , treatment W_i , observed follow-up Y_i and event indicator D_i :

$$Y_i = \min\{T_i, C_i\}, \quad D_i = \mathbf{1}\{T_i \leq C_i\}.$$

If $D_i = 0$ and $Y_i < t$, survival at t is unknown even on the assigned path. Censoring and the missing counterfactual are different sources of missing information. Censored people are not automatically events or survivors at every later horizon.

Identification assumptions to discuss

Well-defined strategies and consistency

Treatment has a specified meaning; the observed outcome follows the assigned strategy. Interference between people must be absent or explicitly modeled.

Conditional exchangeability

Given adequate baseline covariates X , treatment assignment carries no remaining information about potential event times: $\{T(0), T(1)\} \perp W \mid X$. Measuring many variables does not establish this assumption.

Treatment positivity / overlap

For relevant profiles, both treatment choices are possible: $0 < e(x) < 1$, where $e(x) = \mathbb{P}(W = 1 \mid X = x)$ is the **propensity score**. Sparse overlap makes adjustment unstable.

Censoring assumptions and follow-up support

A sufficient working assumption is $T \perp C \mid X, W$, with positive probability of remaining uncensored through the horizons of interest. Relevant common causes of censoring and the event must be handled.

Identification is separate from estimation

These assumptions connect a causal target to the distribution of observed data. An estimator then learns that distribution with finite data and modeling error. Flexible forests cannot repair unmeasured confounding or create support where a treatment is never observed. Generalizing to a different population also needs a defensible sampling or transport argument.

Challenge prompt. A model fits the observed data very well and its confidence interval is narrow. What evidence would still be needed before calling the estimate causal? Name one plausible unmeasured common cause and one overlap check for your proposed study.

CSF assumptions: Cui et al. (2023), [doi:10.1093/jrsssb/qkac001](https://doi.org/10.1093/jrsssb/qkac001). Software target: [grf causal_survival_forest](#) reference.

3 Effects across time and people

Causal survival forests (CSF) estimate conditional average effects with right-censored outcomes. This demo uses the survival-probability target, evaluated at five horizons, then reports a cohort average at each horizon. A **horizon** is a chosen evaluation time; it is not a bin for event times.

What CAST adds

CAST (Causal Analysis for Survival Trajectories) models how the horizon estimates vary over follow-up. In this demo, a quadratic is fitted to the CSF estimates:

$$\hat{\Delta}_{\text{CAST}}(t) = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 t^2.$$

More points drawn along this curve are predictions from that fit, not new observations or additional forest estimates. Keep the original horizon points visible: smoothing can miss a peak, a crossing or another feature.

Optional math: correlated estimates and the band

Let $\hat{\mathbf{d}}$ contain the horizon estimates, B have rows $b(t_j)^\top = (1, t_j, t_j^2)$, and $\hat{\Sigma}$ estimate their joint covariance. With fitting weights Q ,

$$\begin{aligned}\hat{\beta} &= (B^\top Q B)^{-1} B^\top Q \hat{\mathbf{d}}, \\ \hat{V}_\beta &= (B^\top Q B)^{-1} B^\top Q \hat{\Sigma} Q B (B^\top Q B)^{-1}.\end{aligned}$$

The shared people induce dependence across horizons. This implementation forms $\hat{\Sigma}$ from patient-aligned influence scores and applies covariance shrinkage. It uses generalized least squares (GLS), $Q = \hat{\Sigma}^{-1}$, when conditioning and fit checks pass; otherwise it uses weighted least squares (WLS) with inverse-variance diagonal weights. A reported 95% pointwise band at t is

$$b(t)^\top \hat{\beta} \pm 1.96 \sqrt{b(t)^\top \hat{V}_\beta b(t)}.$$

Pointwise does not mean 95% coverage of the whole trajectory at once. The calculation also does not account for all uncertainty from choosing the fit or from a misspecified quadratic. The full procedure's coverage needs separate validation. These implementation choices belong to the demo code; they are not all specified in the original CAST preprint.

Heterogeneity and the meaning of time

Treatment-effect heterogeneity means $\tau(t, x)$ varies across profiles x , on a stated outcome scale and at a stated horizon. Averaging over a target population gives $\Delta(t) = \mathbb{E}_X[\tau(t, X)]$. The website plots cohort averages; it does not display a subgroup analysis or identify Hector's effect.

A falling positive survival difference still favors treatment at that horizon. Following the effect of a *baseline* treatment is different from deciding when to switch or stop it. A fitted peak is not a treatment rule.

CAST framework: [Yang et al. \(2025\)](#). Exact covariance, weighting and band: [versioned R/cast_core.R](#).

4 Class exercises

A. A contrast is not a count of identifiable beneficiaries

In a hypothetical population of 10 people, suppose the average survival effect at one horizon is $+0.20$. Let H count people with $\delta_i(t) = +1$ and L count people with $\delta_i(t) = -1$.

1. Write the relationship among H , L and the average effect.
2. Give two different pairs (H, L) that produce $+0.20$.
3. Does this average tell you that a particular two people were saved? Explain what remains unobserved.

B. Read a frozen website example

These values come from the 2,000-person *reversal* scenario with measured confounding strength 1 and hidden-confounding strength 0. All entries are survival-probability differences in pp, rounded from the frozen export.

Month	Unadjusted	CSF	CAST	Oracle
12	13.49	6.51	6.39	7.51
36	32.35	18.09	10.34	16.38
60	28.56	11.42	9.96	9.15
84	14.51	-0.12	5.28	-1.59
108	6.81	-4.38	-3.72	-5.60

4. At 108 months, which comparisons favor treatment? Interpret the CSF estimate in words. Does its sign tell you when treatment should be stopped?
5. At which of the five horizons is the absolute CAST–oracle gap largest? Compute it. What does this example show about smoothing?

C. Inspect or run the supplied cohort

The kit's `data/demo-cohort.csv` contains **600 synthetic people**, generated with seed 20260905, reversal effects, measured confounding strength 1 and no hidden confounding. It is a distinct sample from the frozen website export. Baseline columns are age, stage, ps, comorb, smoke, sex, ethnicity; treatment is W, follow-up in months is Y, and the event indicator is D. Read category coding in the kit README.

Run `class-lab.R` from the extracted kit as described in its README. It fits Igor's bundled implementation to the cohort. Compare the five horizon estimates with `data/answer-key.csv`, which contains the generating model's probability contrasts for these same profiles. An estimate is not expected to equal the answer key in one finite sample.

Discuss. Describe one baseline imbalance and the fraction censored. Choose a clinically meaningful horizon *before* examining its result. Then state the population, comparison, assumptions and uncertainty in a two-sentence interpretation. Use `data/frozen-horizons.csv` for a no-compute comparison across the website's 24 scenarios.

5 Answer notes and further reading

Answers to the numbered exercises

1. The realized average is $(H - L)/10$, so $H - L = 2$.
2. $(2, 0)$ and $(4, 2)$ both give $+0.20$, with different numbers helped and harmed.
3. No. Even an exact average would not identify the missing potential outcomes. An estimated probability difference also has sampling and modeling uncertainty.
4. At 108 months the unadjusted comparison is positive; CSF, CAST and the oracle are negative. CSF estimates survival under baseline treatment to be 4.38 pp lower than under control for the cohort target. This is a horizon contrast, not a comparison of continuing versus stopping treatment then.
5. At 84 months: $|5.28 - (-1.59)| = 6.87$ pp. The gap at 36 months is 6.04 pp. A smooth fit can miss the generating trajectory even if it looks plausible; it need not improve on the separate horizon estimates.

Facilitator notes for the lab and discussion

The cohort has one row per person, not paired potential outcomes. The answer key averages known model probabilities; do not use it as fitting data. The fraction with $D = 0$ describes recorded censoring, not whether survival is known at every horizon.

An interpretation should state the baseline choices, target profiles, horizon, units and uncertainty. Separate identification failures, such as hidden confounding, from estimation problems, such as poor fit. For a new population, revisit the averaging distribution. For changing treatment over time, define a new longitudinal question and revisit identification.

A compact vocabulary recap

Estimand: the quantity we want to learn. **Identification:** relating that quantity to observed-data distributions under assumptions. **Estimator:** the procedure used to learn it from a sample. **Counterfactual:** an outcome under an alternative specified treatment. **Censoring:** incomplete event-time observation. **Overlap:** support for both treatment choices within relevant profiles. **CSF:** a forest for conditional average effects with censored outcomes. **CAST:** a model for an effect trajectory across horizons. **Pointwise interval:** uncertainty at one specified time. **Heterogeneity:** variation in a conditional average effect across profiles.

Sources and editable materials

- Hernán & Robins (2020). *Causal Inference: What If*. Potential outcomes, identification and generalization.
- Cui et al. (2023). *Estimating heterogeneous treatment effects with right-censored data via causal survival forests*. JRSS B 85:179–211.
- Yang et al. (2025). *CAST: Time-Varying Treatment Effects*. Methods preprint, arXiv:2505.06367.
- Petersen & van der Laan (2014). *Causal models and learning from data*. Epidemiology 25:418–426.
- [grf CSF reference](#) and [Igor Shuryak's demo repository](#): target definitions, source code and implementation details.